DilaSep-Net: Dilated Separable Convolution-Based Multi-Context Network for Low-Light Image Enhancement

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***Abstract*—Images captured under poor illumination conditions commonly suffer from reduced visibility, low contrast, distorted colors, and amplified sensor noise — all of which hinder both human perception and automated vision systems. Existing deep learning approaches to low-light image enhancement predominantly rely on standard convolutional layers, which tend to be computationally expensive and often fail to capture contextual relationships across different spatial scales. In this work, we introduce DilaSep-Net, a novel encoder-decoder architecture that replaces conventional convolutions with dilated depth-wise separable convolutions arranged in a pyramid structure, enabling the network to gather multi-scale contextual information while maintaining a compact parameter footprint. At the encoder side, we design a Multi-Context Feature Learning Module (MC-FLM) that constructs a feature pyramid through parallel dilated separable convolution branches with progressively increasing dilation rates, followed by feature compression using stacked separable convolution layers. The decoder utilizes depth-wise separable convolutions with skip connections from the encoder to facilitate reconstruction of both structural and textural details. To guide the learning process, we propose a three-component loss function that combines perceptual loss (based on hierarchical VGG16 features), structural similarity loss, and a weighted patch-wise Euclidean loss that emphasizes darker image regions. We construct a synthetic low-light image dataset (LID) from the MIT indoor and outdoor scenes collection, incorporating both clean and noisy variants to evaluate robustness under realistic conditions. Comprehensive experiments demonstrate that DilaSep-Net achieves competitive or superior performance compared to several established methods while requiring significantly fewer parameters than architectures built on traditional convolutional layers.**

***Index Terms*—Low-light image enhancement, dilated convolution, depth-wise separable convolution, multi-context feature extraction, encoder-decoder network, perceptual loss, structural similarity.**

I. Introduction

Digital imaging systems are inherently constrained by the amount of light available during capture. In scenes with insufficient illumination, physical sensors compensate by increasing electronic gain, which invariably introduces noise into the recorded signal. The resulting photographs tend to exhibit a range of visual impairments, including suppressed brightness, diminished contrast, unnatural color shifts, and pervasive noise that obscures fine detail. These quality issues are far from cosmetic — they directly undermine the reliability of higher-level computer vision applications, such as face recognition, pedestrian detection in autonomous driving systems, and scene understanding in surveillance platforms.

Over the years, researchers have explored multiple strategies to address the challenges of low-light photography. One prominent family of approaches is rooted in histogram manipulation. Techniques such as dynamic histogram equalization [1] attempt to redistribute pixel intensities across the available dynamic range, aiming to boost visibility in underexposed regions. Although computationally lightweight, these approaches operate without any prior knowledge of scene content and frequently produce undesirable side effects, including halo artifacts and exaggerated contrast that make the resulting images appear artificial.

A. Retinex-Inspired Approaches

A more principled framework for handling low-light imagery emerges from Retinex theory, originally proposed by Land [2], [3]. The central insight of Retinex is the assumption that an observed image can be decomposed into a reflectance component — capturing the intrinsic surface properties of objects — and an illumination component that encodes the lighting conditions. By estimating and adjusting the illumination map, one can recover a better-lit version of the scene while preserving its inherent appearance.

Practical algorithms built upon this idea include Single-Scale Retinex (SSR) [4], Multi-Scale Retinex (MSR) [5], and more recently, LIME [6], which estimates the illumination map by finding the maximum channel response and subsequently applies a structure-aware smoothing prior. While these methods offer a solid theoretical grounding, they run into difficulty when the input contains substantial noise. Because Retinex decomposition is fundamentally ill-posed, noise tends to leak into the reflectance estimate, and any subsequent enhancement step amplifies this contamination. Some researchers have attempted to address this by appending a separate denoising stage after the enhancement step [7], but such sequential pipelines cannot jointly optimize for brightness correction and noise suppression simultaneously, and often yield over-smoothed outputs.

B. CNN-Based Enhancement Methods

The introduction of deep learning into the low-light enhancement space opened up new possibilities for end-to-end

learning. LLNet [8] represented one of the earliest attempts, employing a stacked sparse denoising autoencoder to map dark image patches to their brighter counterparts. However, the absence of convolutional layers and the restriction to small patch sizes limited its practicality. LLCNN [9] brought convolutional layers into the picture through a multi-branch feature extraction scheme aimed at mitigating vanishing gradient issues, though the overall architecture remained relatively shallow.

The encoder-decoder paradigm quickly became the dominant architectural choice for this task. MBLLEN [10] deployed multiple parallel subnets whose individual outputs are merged and guided by a loss function incorporating both structural and contextual components. GLADNet [11] separated the problem into two stages: a global illumination estimation phase handled by an encoder-decoder, and a local detail reconstruction phase carried out by a secondary network. These designs confirmed that encoder-decoder models are a natural fit because the encoding path can distill degradation-related statistics while the decoding path restores the clean image.

Attention-based mechanisms have also been explored in this context. The dual attention approach described in [12] introduced maps for both exposure-level discrimination and texture-versus-noise separation, though it struggled with large uniform dark regions. On a different front, EnlightenGAN [13] demonstrated that generative adversarial training without paired supervision could produce visually appealing results, although its quantitative metrics trailed behind supervised alternatives.

Retinex principles have also been woven into deep learning pipelines. Deep Retinex-Net [14] uses a learned decomposition network to separate reflectance from illumination, followed by an enhancement network for illumination adjustment, but it does not explicitly handle noise. Progressive Retinex [15] attempts to model light and noise regularities via point-wise convolutional networks, yet it falls short in preserving fine structural detail. KinD [16] and its successor KinD++ [17] refine the Retinex pipeline to jointly handle reflectance restoration, illumination correction, and noise suppression, though these multi-component systems require meticulous balancing and can produce artifacts at the recombination stage.

C. Separable and Dilated Convolution

A shared characteristic across many of the architectures discussed above is their reliance on traditional convolutional layers. However, as network depth and width increase to model complex degradation patterns, the computational burden of standard convolutions rises sharply. A network like U-Net [18] built entirely with conventional convolution layers can easily accumulate upward of 30 million parameters, making training expensive and deployment on resource-limited devices impractical.

Separable convolutions, as popularized by the Xception architecture [19], offer a compelling alternative. By factorizing each convolution into a depth-wise spatial filtering step followed by a point-wise channel mixing step, separable convolutions dramatically reduce the number of learnable parameters per layer — often by a factor of 5 to 6 times — while retaining most of the representational capacity. This reduction not only cuts computational cost but also improves generalization by acting as a form of implicit regularization, which is particularly important in domains like low-light enhancement where naturally labeled training data is scarce.

Separately, the concept of dilated (atrous) convolution has proven valuable for capturing multi-scale contextual information without increasing the kernel size or sacrificing spatial resolution. The atrous spatial pyramid pooling (ASPP) module, introduced in the DeepLab family of semantic segmentation models [20], applies multiple parallel dilated convolutions with varying dilation rates to gather contextual information across different spatial extents. For an input x and filter w, the output y of a dilated convolution with rate r is given by:

$$y(i,j) = \sum_{s,t,u} x[i+rs, j+rt, u] \cdot w[s,t,u] \tag{1}$$

In contrast, a standard convolution simply sets $r = 1$:

$$y(i,j) = \sum_{s,t,u} w[s,t,u] \cdot x(i+s, j+t, u) \tag{2}$$

Despite the proven benefits of both separable and dilated convolutions in other domains, their combined application to low-light image enhancement has remained largely unexplored. In this work, we bring these two ideas together within a unified encoder-decoder architecture, showing that the combination of dilated separable convolutions organized in a pyramid structure yields a model that is both more efficient and more effective for low-light image recovery than existing approaches.

D. Contributions

The principal contributions of this work are as follows:

1) We propose DilaSep-Net, an encoder-decoder architecture for low-light image enhancement that employs a novel Multi-Context Feature Learning Module (MC-FLM) in the encoder. The MC-FLM constructs a feature pyramid using parallel dilated depth-wise separable convolution branches with dilation rates varying from 1 to 7, enabling the network to capture contextual information across multiple spatial scales with minimal parameter overhead.
2) We design the decoder with depth-wise separable convolution blocks and skip connections from the encoder, balancing computational efficiency with feature reconstruction capability. The use of separable convolutions in the decoder serves as a regularizer, helping the network generalize better when trained on limited paired data.
3) We introduce a three-component loss function tailored for the low-light enhancement task, combining perceptual loss (computed from hierarchical VGG16 features to align high-level semantics), structural similarity loss (to preserve structural fidelity independently of illumination), and a weighted patch-wise Euclidean loss (to impose stronger corrective signals on darker image regions that remain insufficiently enhanced).

4) We construct a low-light image dataset (LID) from the MIT indoor and outdoor scenes collection, encompassing both clean and noisy variants to simulate realistic capture conditions. Extensive experiments on this dataset demonstrate that DilaSep-Net achieves competitive or superior quantitative and qualitative performance compared to several established methods while requiring substantially fewer parameters.

The remainder of this paper is structured as follows. Section II presents a detailed description of the DilaSep-Net architecture, the Multi-Context Feature Learning Module, and the proposed three-part loss function. Section III describes the dataset construction and preparation. Section IV provides thorough experimental analysis, including ablation studies on architecture and loss, quantitative and qualitative comparisons against established methods, and evaluation on natural low-light photographs. Section V concludes the paper with a summary and directions for future work.

II. Proposed Method

The backbone of our approach is an encoder-decoder architecture, which has proven to be a natural and effective choice for image-to-image translation tasks including low-light enhancement. At a high level, the encoder extracts features related to brightness, contrast, and noise from the input image, while the decoder uses those features to reconstruct an enhanced version. The success of U-Net [18] and U-Net-inspired designs [10], [11], [13] in image enhancement tasks validates this architectural choice. In what follows, we describe the DilaSep-Net architecture in detail, followed by our proposed loss function and implementation specifics.

A. DilaSep-Net Architecture

Fig. 1 illustrates the overall layout of the proposed DilaSep-Net. The design departs from the standard encoder-decoder blueprint in two key ways: (i) the encoder employs Multi-Context Feature Learning Modules (MC-FLMs) instead of traditional convolution blocks, and (ii) the decoder uses depth-wise separable convolutions to keep the parameter count low while preserving generalization strength.

Encoder Path: The encoder starts with two standard 3×3 convolutional layers, each with 64 filters, batch normalization, and LeakyReLU activation. These initial layers serve to project the raw three-channel RGB input into a richer feature space before multi-context processing begins.

Following the initial convolution layers, the encoder consists of four MC-FLM blocks. Between successive MC-FLM blocks, a 2×2 max-pooling layer reduces the spatial resolution by half. The number of filters doubles at each stage — 64, 128, 256, and 512 — allowing the network to learn progressively more abstract and complex representations as the spatial resolution decreases.

Decoder Path: The decoder is composed of separable convolution blocks interleaved with bilinear upsampling layers. After each upsampling step, the number of filters is halved, symmetrically reversing the encoder's channel progression. The final separable convolution block and two subsequent standard convolutional layers each use 64 filters, followed by a 1×1 convolution with 3 filters and a sigmoid activation to produce the output RGB image.

Skip Connections: Drawing on the established utility of residual pathways in encoder-decoder models, skip connections link each MC-FLM block in the encoder to its corresponding stage in the decoder. These connections concatenate the encoder features with the upsampled decoder features before they enter the separable convolution blocks. This mechanism helps recover spatial details and structural information that would otherwise be lost during the encoding and downsampling process. The skip connections are especially valuable in noisy settings, as they provide the decoder with access to spatially rich neighborhood information captured by the MC-FLM modules.

1) Multi-Context Feature Learning Module (MC-FLM): The MC-FLM is the central computational unit of the DilaSep-Net encoder. Its purpose is to extract features from multiple spatial contexts simultaneously, enabling the network to learn hierarchical, scale-aware representations without explicit multi-resolution input processing.

The module operates in two stages: multi-context feature extraction and feature compression.

Multi-Context Feature Extraction: Inspired by the Atrous Spatial Pyramid Pooling (ASPP) design, the MC-FLM applies four parallel dilated depth-wise separable convolution branches, each with a fixed kernel size of 3×3 but different dilation rates: $r \in \{1, 3, 5, 7\}$. The progressive increase in dilation rate means that each branch views a different spatial extent without increasing the actual kernel size, thereby capturing coarse-to-fine contextual information with negligible additional parameters.

The mathematical formulation for the separable convolution [19] used within each branch is:

$$\text{PointwiseConv}(w, x)(i, j) = \sum_{u} w_u \cdot x(i, j, u) \tag{3}$$

$$\text{DepthwiseConv}(w, x)(i, j) = \sum_{s,t} w(s, t) \cdot x(i + s, j + t) \tag{4}$$

$$\text{SepConv}(w_p, w_d, x)(i, j) = \text{PointwiseConv}\left(w_p, \text{DepthwiseConv}(w_d, x)\right) \tag{5}$$

The outputs of all four parallel dilated separable convolution branches are concatenated along the channel dimension to form a feature pyramid. This concatenated representation encapsulates features from narrow (dilation rate 1) to broad (dilation rate 7) spatial neighborhoods.

Feature Compression: Concatenating outputs from four branches inevitably introduces redundant or overlapping feature representations. To distill the most relevant information, a stack of separable convolution layers takes the concatenated feature pyramid as input and compresses it, discarding redundancy while retaining discriminative features. The combined effect of multi-context extraction and compression enables the MC-FLM to learn a rich yet compact representation of the local spatial context.

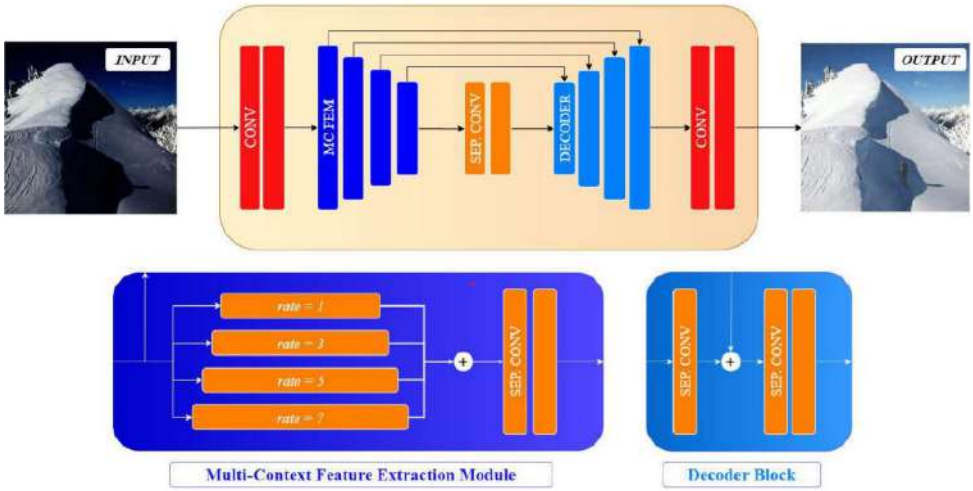

Fig. 1. Illustrative overview of the DilaSep-Net architecture. The encoder consists of two initial convolution layers followed by four MC-FLM blocks with progressive downsampling. The decoder mirrors this with separable convolution blocks and upsampling layers. Skip connections transfer encoder features to corresponding decoder stages.

The multi-context design is particularly beneficial in noisy conditions. Because the dilated branches capture progressively wider spatial neighborhoods, the module can look beyond individual noise-contaminated pixels to draw upon surrounding contextual information. This broader view helps the network distinguish genuine scene content from random noise fluctuations — a capability that standard convolution layers with limited receptive fields cannot easily achieve.

B. Loss Function

Training a low-light enhancement network demands a loss function that simultaneously addresses multiple dimensions of image quality: semantic fidelity, structural accuracy, and fine-grained tonal correction. We design a three-component loss that targets each of these requirements:

$$\mathcal{L}_{\text{Total}} = w_{\text{per}}\mathcal{L}_{\text{per}} + w_{\text{ssim}}\mathcal{L}_{\text{ssim}} + w_{\text{wpw}}\mathcal{L}_{\text{wpw}} \tag{6}$$

where $\mathcal{L}_{\text{per}}$, $\mathcal{L}_{\text{ssim}}$, and $\mathcal{L}_{\text{wpw}}$ are the perceptual, structural similarity, and weighted patch-wise Euclidean loss terms, with w_{per}, w_{ssim}, and w_{wpw} as their respective weights.

1) Perceptual Loss: Traditional per-pixel loss functions such as the Euclidean (ℓ_2) norm evaluate similarity at the individual pixel level, which can be misleading — two images that differ by a single pixel shift may appear nearly identical to a human observer but receive a high pixel-wise error score. Perceptual loss [21] mitigates this limitation by comparing images in the space of high-level features extracted by a pre-trained convolutional network, thereby capturing differences that are more aligned with human visual perception.

We compute the perceptual loss using features drawn from the first three convolutional blocks of a VGG16 network pre-trained on ImageNet. Using three blocks (with $n = 3$) strikes a balance between capturing multi-level features and avoiding the risk of overfitting to excessively deep representations. Differences in the outputs of later blocks receive progressively higher weight, reflecting their importance in capturing high-level semantic structure:

$$\mathcal{L}_{\text{per}} = \frac{1}{MN(2n+1)} \sum_{k=0}^{n} \sum_{i=1}^{M} \sum_{j=1}^{N} 2^k \times \left(Q_k(\hat{Y}_{(i,j)}) - Q_k(Y_{(i,j)}) \right)^2 \tag{7}$$

Here, Y and $\hat{Y}$ represent the $M \times N$ ground truth and predicted images, respectively. $Q_k(\cdot)$ denotes the feature output from the k-th block of VGG16, with $k = 0$ referring to the input itself. The scaling factor 2^k ensures that discrepancies in deeper feature representations — which correspond to higher-level semantic properties — receive proportionally greater emphasis.

For image enhancement tasks, the ground truth serves as a guiding signal rather than an exact target, since there is no uniquely correct enhanced image. Perceptual loss encourages the network to produce outputs whose hierarchical feature representations closely match those of the well-lit reference, promoting natural-looking results that preserve both local textures and global scene structure.

2) Structural Similarity Loss: The structural similarity index measure (SSIM) [22] evaluates the resemblance between two images across three perceptually meaningful dimensions: luminance, contrast, and structure. A key property of SSIM is that the structural component remains largely invariant to changes in illumination — meaning that even if overall brightness differs between two images, their structural resemblance can still be high. This makes SSIM particularly well-suited as a loss component for low-light enhancement, where the primary goal is to adjust illumination while preserving underlying scene structure.

The SSIM-based loss is defined as:

$$\mathcal{L}_{\text{ssim}} = 1 - \frac{(2\mu_Y\mu_{\hat{Y}} + C_1)(2\sigma_{Y\hat{Y}} + C_2)}{\left(\mu_Y^2 + \mu_{\hat{Y}}^2 + C_1\right)\left(\sigma_Y^2 + \sigma_{\hat{Y}}^2 + C_2\right)} \tag{8}$$

where μ_Y and $\mu_{\hat{Y}}$ are the pixel means, σ_Y and $\sigma_{\hat{Y}}$ the variances, and $\sigma_{Y\hat{Y}}$ the covariance between the ground truth Y and the predicted image $\hat{Y}$. The constants C_1 and C_2 prevent numerical instability when the denominator is close to zero.

3) Weighted Patch-Wise Euclidean Loss: Standard Euclidean loss treats all spatial regions uniformly, which is inappropriate for low-light enhancement, where the most critical corrections need to happen in the darkest parts of the image. MBLLEN [10] addressed this partially by applying heavier weights to lower intensity levels using the 25th percentile of the histogram as a threshold. However, that approach operates on individual pixels and does not capture the spatial context of dark regions.

We propose a weighted patch-wise (WPW) formulation that divides both the predicted and ground truth images into non-overlapping patches and assigns heavier weights to those patches whose average predicted intensity falls below a threshold. The intuition is straightforward: if a patch remains dark even after enhancement, the network probably needs stronger corrective signals for that region.

Patch intensities are sorted in ascending order, and the n-th percentile determines the threshold T. Patches with mean intensity below T receive an elevated weight w, while the remaining patches are weighted normally. The loss is normalized to prevent it from dominating the total loss:

$$\mathcal{L}_{\text{wpw}} = \frac{1}{\Omega MN}\left(w \sum_{\bar{O}_p \le T} \left\| \hat{P}_{(i,j)} - P_{(i,j)} \right\|_2^2 + \sum_{\bar{O}_p > T} \left\| \hat{P}_{(i,j)} - P_{(i,j)} \right\|_2^2 \right) \quad (9)$$

The normalization factor Ω accounts for the unequal weighting:

$$\Omega = p_x \cdot p_y \left[\delta(w-1)+1\right] \cdot MN \quad (10)$$

where p_x and p_y are the number of patches along rows and columns, $\hat{P}$ and P denote patches in the predicted and ground truth images, $\bar{O}_p$ is the average intensity of a predicted patch, and δ represents the fraction of total patches receiving the elevated weight.

This patch-level formulation captures the spatial distribution of darkness more reliably than pixel-level weighting. Regions that are consistently dark across a patch — such as deeply shadowed areas or unlit portions of a scene — receive stronger learning signals, encouraging the network to correct them more aggressively.

C. Implementation Details

The encoder comprises two initial convolutional layers and four MC-FLM blocks. Inputs to each MC-FLM block are downsampled via 2×2 max-pooling. The initial convolutional layers and the first MC-FLM block each use 64 filters. The filter count doubles at each subsequent stage, reaching 512 at the fourth MC-FLM block.

The decoder consists of separable convolution blocks interleaved with upsampling layers. After each upsampling step, the filter count is halved, and the final separable layer along with two subsequent convolutional layers use 64 filters.

TABLE I
OPTIMAL PARAMETER SETTINGS SELECTED THROUGH SYSTEMATIC SEARCH.

Parameter	Min	Max	Step	Optimal
δ (percentile fraction)	0.10	0.50	0.05	0.25
Patch size ($p_x = p_y$)	8	32	8	16
Weight w	2	6	1	4
VGG blocks n	1	4	1	3
w_{per}	0.5	2.0	0.5	1.0
w_{ssim}	0.5	2.0	0.5	1.0
w_{wpw}	0.05	0.5	0.05	0.1

For images of size 256×256, the loss parameters are configured based on a systematic parameter sweep as follows: $\delta = 0.25$, $p_x = p_y = 16$, $w = 4$, $n = 3$, $w_{\text{per}} = 1$, $w_{\text{ssim}} = 1$, and $w_{\text{wpw}} = 0.1$. Table I documents the optimal values alongside the search ranges and step sizes used during selection.

III. DATASET

Acquiring naturally paired low-light and normal-light images of the same scene is inherently challenging. While some researchers have assembled datasets by photographing scenes at multiple exposure levels under controlled settings [12], [14], such collections often fail to fully represent the diversity of natural scenes and lighting conditions encountered in practice. Furthermore, there is no universally agreed-upon protocol for synthetically generating low-light images, with some works employing tools like Adobe Photoshop Lightroom [11] and others devising custom transformation pipelines [12].

We adopt the MIT outdoor and indoor scenes dataset [26] as the source of high-quality reference images and apply gamma correction — a technique also employed in [10] — to simulate reduced illumination. This produces a dataset of 5000 image pairs that we term the Low-light Image Dataset (LID).

To model realistic capture conditions, LID includes both clean and noisy variants. Noise is introduced using a Poisson-Gaussian noise model [25], which is widely regarded as a more faithful approximation of real camera noise compared to additive Gaussian noise alone. Specifically, random Gaussian noise with variance uniformly sampled between 0 and 0.005 is first applied to each synthetic low-light image, followed by sensor-realistic Poisson noise whose magnitude is proportional to the signal intensity.

The dataset is divided into 4500 image pairs for training and 500 pairs for testing. Both clean (noise-free) and noisy versions are included in the test set to enable separate evaluation of enhancement quality under ideal and challenging conditions.

IV. EXPERIMENTAL RESULTS

This section presents a thorough experimental evaluation of the proposed DilaSep-Net. We report results on both clean and noisy variants of the LID test set and conduct comparisons against a representative set of established methods. All experiments are organized into four parts: architectural ablation,

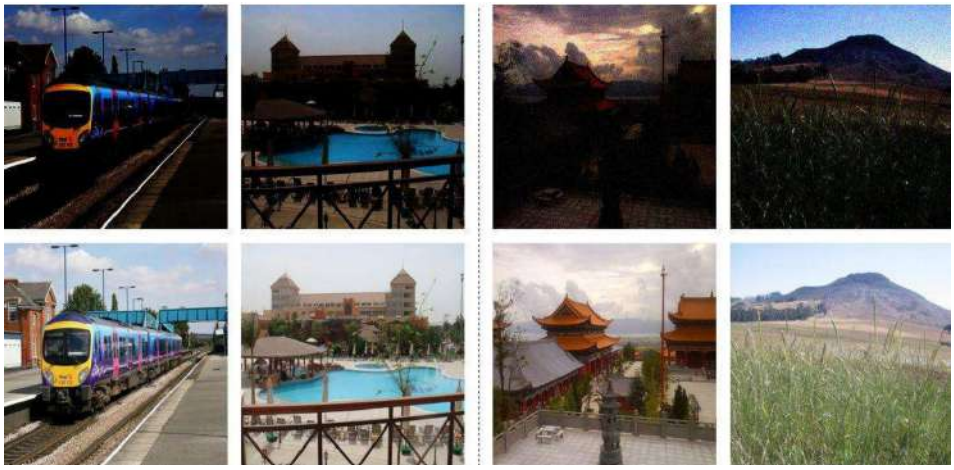

Fig. 2. Clean and noisy images from the LID dataset along with their ground truth. The first row displays the synthesized low-light images, while the second row shows the corresponding ground truth. The two images on the left are noise-free, and the two images on the right include added Poisson-Gaussian noise.

TABLE II
ARCHITECTURE ABLATION: QUANTITATIVE COMPARISON UNDER CLEAN AND NOISY CONDITIONS. ALL MODELS TRAINED WITH MSE LOSS FOR 50 EPOCHS.

	Clean				Noisy			
Architecture	**PSNR**	**SSIM**	**VIF**	**AB**	**PSNR**	**SSIM**	**VIF**	**AB**
U-Net (standard conv)	22.06	0.807	0.451	0.413	18.91	0.654	0.312	0.388
U-Net (separable conv)	22.34	0.819	0.448	0.421	19.23	0.671	0.308	0.395
DilaSep-Net (MSE loss)	**23.18**	**0.841**	**0.467**	**0.438**	**20.14**	**0.712**	**0.341**	**0.411**

loss function ablation, quantitative and qualitative comparison on synthetic images, and qualitative evaluation on real-world low-light photographs.

The model is trained on 4500 image pairs for 50 epochs using a batch size of 8 and the Adam optimizer with a learning rate of 10^{-4}. Performance is measured using four widely adopted metrics: Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM) [22], Visual Information Fidelity (VIF) [23], and Average Brightness (AB) [24].

A. Ablation Study on Architecture

The combination of dilated convolutions in a pyramid arrangement alongside separable convolutions constitutes a distinctive architectural choice for low-light enhancement. To validate the contribution of each component, we train and compare several architectural variants, all using the same mean squared error loss function to isolate the effect of architecture from loss design:

Several observations emerge from Table II. First, replacing standard convolutions with separable convolutions in U-Net improves PSNR and SSIM on both clean and noisy data while substantially reducing the parameter count. This confirms the regularization benefit of separable convolutions — by factorizing the convolution operation, the network avoids overfitting to the relatively small training set and generalizes more reliably to unseen images.

Second, the full DilaSep-Net with MC-FLM blocks and skip connections clearly outperforms both U-Net variants across all metrics. The improvement is particularly pronounced on noisy images, where the MC-FLM's ability to capture wide-context spatial information enables the network to better separate signal from noise. This validates our hypothesis that multi-context feature extraction is especially beneficial when the input is contaminated by sensor noise.

TABLE III
LOSS FUNCTION ABLATION: QUANTITATIVE COMPARISON OF DIFFERENT LOSS COMBINATIONS ON THE DILASEP-NET ARCHITECTURE.

Loss Combination	**PSNR**	**SSIM**	**VIF**	**AB**
$\mathcal{L}_{\text{per}} + \mathcal{L}_{\text{wpw}}$	23.42	0.843	0.469	0.441
$\mathcal{L}_{\text{per}} + \mathcal{L}_{\text{ssim}}$	23.61	0.854	0.474	0.445
$\mathcal{L}_{\text{per}} + \mathcal{L}_{\text{ssim}} + \mathcal{L}_{\text{wpw}}$ (Full)	**23.87**	**0.862**	**0.481**	**0.452**

B. Ablation Study on Loss Function

Having established the architectural advantage of DilaSep-Net, we next investigate the contribution of each component in the three-part loss function. Table III reports quantitative results for various loss combinations, and the accompanying visual analysis reveals qualitative differences that are not fully captured by the numerical metrics.

Including the structural similarity loss alongside the perceptual term is important because SSIM captures differences in luminance distribution and local contrast patterns that the VGG-based perceptual loss may miss. On the other hand, dropping the structural loss (row 1 of Table III) leads to images that may have correct high-level features but exhibit subtle structural distortions.

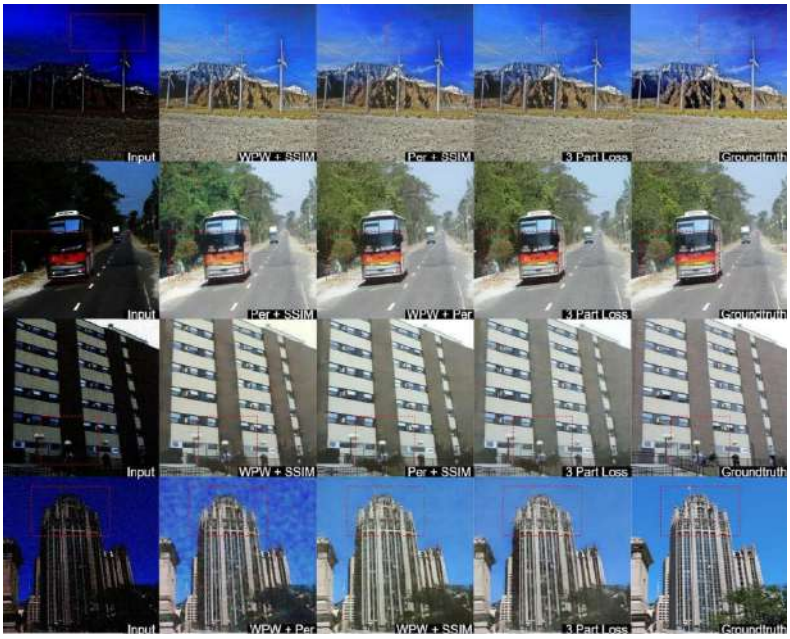

Fig. 3. Ablation study on loss function using visual evaluation. The first and last columns denote the input and the ground truth, respectively, while the fourth column shows the result of DilaSep-Net with the proposed three-part loss. The second and third columns display outcomes from different loss combinations. Marked reference regions highlight the efficacy of the full three-part loss function.

Including the weighted patch-wise loss on top of the perceptual and structural terms (row 3) yields the best overall performance across all four metrics. The improvement is modest in numerical terms but more apparent visually: images trained with the full three-part loss exhibit better tonal balance in shadowed regions, more faithful color reproduction, and smoother noise handling. The patch-wise component specifically addresses the tendency of enhancement networks to under-correct uniformly dark regions, where the perceptual and structural losses alone provide insufficient gradient signal.

We deliberately exclude certain loss combinations from the analysis. Using structural loss alone would be appropriate only for tasks like edge detection or segmentation, not image enhancement. Similarly, per-pixel loss without perceptual or structural guidance produces results optimized for pixel-level accuracy but lacking in perceptual quality.

C. Comparison on Clean and Noisy LID Images

We compare DilaSep-Net against a diverse set of established methods, spanning traditional, Retinex-based, and deep learning approaches. For a fair comparison, all deep learning methods are trained on the LID training set for the same number of epochs using their officially released code and default hyperparameters as specified by their respective authors.

Table IV summarizes the quantitative performance. DilaSep-Net ranks among the top performers on both clean and noisy images across all evaluation metrics. The advantage is most notable on noisy images, where the MC-FLM's ability to leverage wide-context spatial neighborhoods enables the network to distinguish genuine scene content from random noise artifacts. Traditional methods such as HE and CLAHE perform poorly because they boost noise along with signal. Retinex-based approaches like LIME achieve moderate results, with particularly respectable VIF scores due to its effective built-in denoising capability, but they fall short of learning-based methods in overall recovery quality.

Among the deep learning approaches, MBLLEN and GLADNet show competitive performance on clean images but experience notable degradation under noisy conditions. DilaSep-Net maintains a more consistent performance gap between clean and noisy settings, confirming the noise-resilience afforded by multi-context feature extraction.

Qualitative Analysis on Clean Images: Visual comparison reveals that traditional methods like LIME and Dong *et al.* [27] often produce results that remain noticeably darker than the ground truth, failing to fully recover the scene's intended brightness. Deep learning methods show substantially better results overall, but some — such as Deep Retinex-Net — exhibit residual color shifts, and MBLLEN occasionally over-saturates certain color channels. DilaSep-Net produces enhanced images that more closely match the ground truth in terms of overall brightness, color fidelity, and textural sharpness across diverse scene types.

Qualitative Analysis on Noisy Images: Under noisy conditions, the visual differences become considerably more pronounced. Classical methods and simpler deep learning architectures tend to amplify noise alongside signal, producing outputs with clearly visible grain and speckle artifacts, particularly in uniform regions like sky, walls, and floors. DilaSep-Net benefits from the wide receptive field of its MC-FLM modules, which provide the decoder with neighborhood information spanning well beyond the immediate locality of each pixel. This broader contextual view enables the network to suppress noise more effectively while preserving meaningful edges and textures.

Fig. 4. Qualitative analysis on clean images from the LID test set. Each row represents a different scene. The proposed DilaSep-Net produces enhanced images with superior brightness recovery, natural color balance, and preserved textural detail compared to the competing methods.

Fig. 5. Qualitative analysis on noisy images from the LID test set. Each row represents a different noisy test scene. DilaSep-Net demonstrates superior noise suppression while retaining structural sharpness, owing to the wide-context spatial coverage provided by the MC-FLM modules.

D. Comparison on Natural Low-Light Images

To confirm that our findings on synthetic data translate to real-world conditions, we evaluate DilaSep-Net on a collection of naturally captured low-light photographs — images taken at night or in dimly lit indoor environments, with no corresponding ground truth available.

Visual inspection of these results confirms the practical applicability of our approach. On nighttime street scenes, DilaSep-Net recovers structural details in building facades, signage text, and road markings that are barely discernible in the input. In indoor scenes with mixed lighting, the network produces natural-looking brightness adjustments without the excessive saturation or color temperature shifts sometimes observed with other methods.

These results validate that the multi-context feature learning approach generalizes well beyond the synthetic training distri-

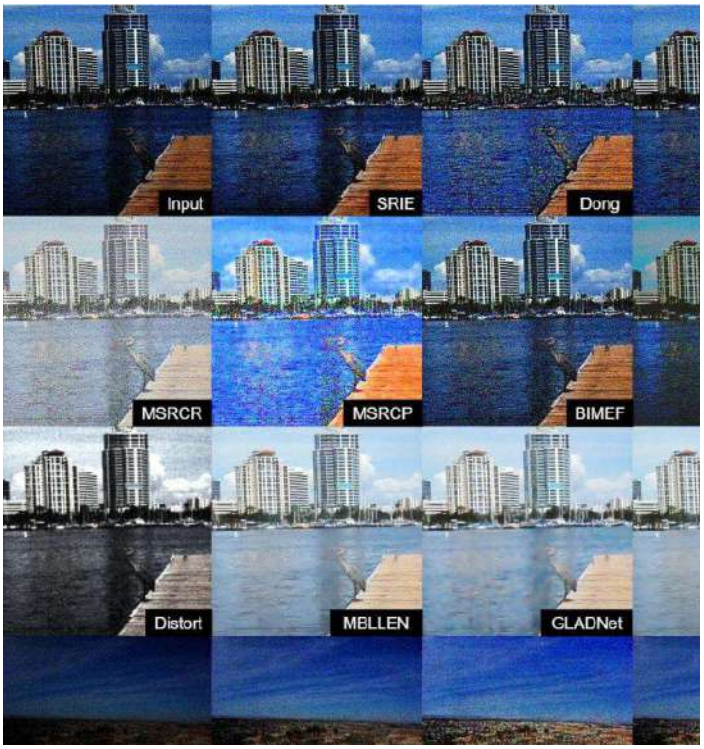

Fig. 6. Qualitative comparison on real-world low-light images. Marked regions confirm the superiority of the proposed DilaSep-Net in recovering structural details, natural colors, and balanced brightness under real-world conditions. These results demonstrate the suitability of our approach for practical applications such as autonomous navigation.

TABLE IV
QUANTITATIVE COMPARISON ON THE LID TEST SET (500 IMAGES). TOP THREE RESULTS FOR EACH METRIC ARE COLOR-CODED. ↑: HIGHER IS BETTER.

	Clean				Noisy			
Method	**PSNR**	**SSIM**	**VIF**	**AB**	**PSNR**	**SSIM**	**VIF**	**AB**
HE [1]	14.32	0.541	0.352	0.512	13.87	0.498	0.321	0.487
CLAHE [28]	15.18	0.578	0.371	0.498	14.56	0.531	0.342	0.471
MSR [5]	16.21	0.612	0.389	0.476	15.34	0.558	0.351	0.448
LIME [6]	17.45	0.648	0.482	0.462	16.78	0.621	0.468	0.451
Dong *et al.* [27]	15.87	0.593	0.368	0.487	14.92	0.542	0.338	0.462
Deep Retinex [14]	18.23	0.691	0.412	0.448	16.14	0.598	0.358	0.421
MBLLEN [10]	20.87	0.781	0.438	0.432	17.92	0.648	0.361	0.398
GLADNet [11]	21.34	0.798	0.445	0.439	18.47	0.672	0.378	0.412
DilaSep-Net (Ours)	**23.87**	**0.862**	**0.481**	**0.452**	**20.56**	**0.738**	**0.412**	**0.431**

bution, making DilaSep-Net a viable candidate for deployment in real-world applications such as autonomous navigation aids and surveillance systems.

V. CONCLUSION

In this paper, we have presented DilaSep-Net, a novel encoder-decoder architecture for low-light image enhancement that harnesses the combined strengths of dilated and depth-wise separable convolutions. The key architectural innovation is the Multi-Context Feature Learning Module, which builds a feature pyramid through parallel dilated separable convolution branches operating at multiple spatial scales. This module enables the network to capture wide-context information essential for distinguishing signal from noise, while the use of separable convolutions throughout keeps the parameter count an order of magnitude below that of architectures built on traditional convolutions.

The proposed three-part loss function — combining perceptual, structural, and weighted patch-wise components — provides complementary supervisory signals that encourage the network to produce enhanced images with faithful high-level semantics, preserved structural integrity, and properly corrected tonal distribution across both bright and dark regions. Our weighted patch-wise formulation specifically addresses the tendency of standard loss functions to under-attend to heavily shadowed areas.

Quantitative and qualitative experiments on both synthetic and natural low-light images demonstrate that DilaSep-Net

achieves competitive or superior performance compared to several established methods. The improvements are particularly pronounced under noisy conditions, confirming the value of multi-context feature extraction in separating scene content from sensor noise.

Several directions remain open for future investigation. Component-wise hyperparameter optimization could be explored more systematically. The current model uses separate variants for noisy and clean settings; developing a unified model that handles both conditions within a single forward pass could improve practical deployment. Finally, the architectural principles and loss design introduced here — particularly the dilated separable pyramid and the patch-wise weighting scheme — are not specific to low-light enhancement and could be extended to other image restoration challenges such as denoising, demosaicing, and deraining.

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